SQL PROJECT – SYED KHAJA FAKHRUDDIN-

CREATING A DATABASE:

SQL DATA ANALYSIS;

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– Create Customers Table
CREATE TABLE Customers (
    customer_id INT PRIMARY KEY,
    customer_name VARCHAR(100),
    customer_email VARCHAR(100),
    customer_city VARCHAR(100)
);

– Create Categories Table
CREATE TABLE Categories (
    category_id INT PRIMARY KEY,
    category_name VARCHAR(100)
);

– Create Products Table with category_id as foreign key from the beginning
CREATE TABLE Products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    price DECIMAL(10, 2),
    category_id INT,
    FOREIGN KEY (category_id) REFERENCES Categories(category_id)
);
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– Create Orders Table
CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    order_date DATE,
    customer_id INT,
    total_amount DECIMAL(10, 2),
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
);

– Create OrderDetails Table
CREATE TABLE OrderDetails (
    order_detail_id INT PRIMARY KEY,
    order_id INT,
    product_id INT,
    quantity INT,
    price DECIMAL(10, 2),
    FOREIGN KEY (order_id) REFERENCES Orders(order_id),
    FOREIGN KEY (product_id) REFERENCES Products(product_id)
);

– Create CityRegionMapping Table
CREATE TABLE CityRegionMapping (
    city_name VARCHAR(100) PRIMARY KEY,
    region VARCHAR(50)
);

– Insert Sample Data into Categories Table
INSERT INTO Categories (category_id, category_name) VALUES
(1, 'Electronics'),
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(2, 'Appliances'),
(3, 'Personal Care'),
(4, 'Furniture'),
(5, 'Fitness'),
(6, 'Sports');

-- Insert Sample Data into Customers Table
INSERT INTO Customers (customer_id, customer_name, customer_email, customer_city) VALUES
(1, 'Alice Johnson', 'alice.johnson@example.com', 'New York'),
(2, 'Bob Smith', 'bob.smith@example.com', 'Los Angeles'),
(3, 'Charlie Brown', 'charlie.brown@example.com', 'Chicago'),
(4, 'David Williams', 'david.williams@example.com', 'Houston'),
(5, 'Eve Davis', 'eve.davis@example.com', 'Phoenix'),
(6, 'Frank Harris', 'frank.harris@example.com', 'Philadelphia'),
(7, 'Grace Moore', 'grace.moore@example.com', 'San Antonio'),
(8, 'Hank Thompson', 'hank.thompson@example.com', 'San Diego'),
(9, 'Ivy Clark', 'ivy.clark@example.com', 'Dallas'),
(10, 'Jack White', 'jack.white@example.com', 'San Jose');

-- Insert Sample Data into Products Table
INSERT INTO Products (product_id, product_name, price, category_id) VALUES
(1, 'Laptop', 999.99, 1),
(2, 'Smartphone', 799.99, 1),
(3, 'Headphones', 199.99, 1),
(4, 'Camera', 599.99, 1),
(5, 'Refrigerator', 1200.00, 2);

-- Insert Sample Data into Orders Table
INSERT INTO Orders (order_id, order_date, customer_id, total_amount) VALUES
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(1, '2023-08-01', 1, 1249.98),
(2, '2023-08-02', 2, 349.99),
(3, '2023-08-03', 3, 549.99),
(4, '2023-08-04', 4, 799.00),
(5, '2023-08-05', 5, 1234.50);

– Insert Sample Data into OrderDetails Table
INSERT INTO OrderDetails (order_detail_id, order_id, product_id, quantity, price) VALUES
(1, 1, 1, 1, 999.99),
(2, 2, 2, 1, 799.99),
(3, 3, 3, 2, 199.99),
(4, 4, 4, 1, 599.99),
(5, 5, 5, 1, 1200.00);

– Populate CityRegionMapping Table with Sample Data
INSERT INTO CityRegionMapping (city_name, region) VALUES
('New York', 'Northeast'),
('Los Angeles', 'West'),
('Chicago', 'Midwest'),
('Houston', 'South'),
('Phoenix', 'West'),
('Philadelphia', 'Northeast'),
('San Antonio', 'South'),
('San Diego', 'West'),
('Dallas', 'South'),
('San Jose', 'West');

– 1. Retrieve Customer Information
SELECT customer_name, customer_city
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FROM Customers;

– 2. List All Orders
SELECT order_id, order_date, total_amount
FROM Orders;

– 3. Find Products in a Specific Category
SELECT p.product_name, p.price, c.category_name
FROM Products p
JOIN Categories c ON p.category_id = c.category_id
WHERE c.category_name = 'Electronics';

– 4. Total Sales by Customer
SELECT c.customer_name, SUM(o.total_amount) as total_spent
FROM Customers c
JOIN Orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.customer_name
ORDER BY total_spent DESC;

– 5. Count Orders Per City
SELECT c.customer_city, COUNT(o.order_id) as order_count
FROM Customers c
LEFT JOIN Orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_city
ORDER BY order_count DESC;

– 6. Average Quantity per Product
SELECT p.product_name, AVG(od.quantity) as avg_quantity
FROM Products p
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JOIN OrderDetails od ON p.product_id = od.product_id
GROUP BY p.product_id, p.product_name
ORDER BY avg_quantity DESC;

– 7. Top 5 Most Expensive Products
SELECT product_name, price
FROM Products
ORDER BY price DESC
LIMIT 5;
– 8. Customer Spending Rank
SELECT
    c.customer_name,
    SUM(o.total_amount) AS total_spent,
    RANK() OVER (ORDER BY SUM(o.total_amount) DESC) AS spending_rank
FROM Customers c
JOIN Orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.customer_name
ORDER BY total_spent DESC;

– 9. Customers with Above-Average Orders
SELECT
    c.customer_name,
    o.order_id,
    o.total_amount
FROM Customers c
JOIN Orders o ON c.customer_id = o.customer_id
WHERE o.total_amount > (SELECT AVG(total_amount) FROM Orders);

– 10. Products with Highest Sales in Each Category
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WITH CategoryProductSales AS (
    SELECT
        c.category_name,
        p.product_name,
        SUM(od.quantity * od.price) AS total_sales,
        RANK() OVER (PARTITION BY c.category_id ORDER BY SUM(od.quantity * od.price)
DESC) AS sales_rank
    FROM Categories c
    JOIN Products p ON c.category_id = p.category_id
    JOIN OrderDetails od ON p.product_id = od.product_id
    GROUP BY c.category_id, p.product_id
)
SELECT
    category_name,
    product_name,
    total_sales
FROM CategoryProductSales
WHERE sales_rank = 1;

– 11. Customers with Multiple Orders
SELECT
    c.customer_name,
    COUNT(o.order_id) AS order_count
FROM Customers c
JOIN Orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.customer_name
HAVING COUNT(o.order_id) > 1;

– 12. Products Ordered by Each Customer
SELECT
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    c.customer_name,
    p.product_name,
    SUM(od.quantity) AS total_quantity_ordered
FROM Customers c
JOIN Orders o ON c.customer_id = o.customer_id
JOIN OrderDetails od ON o.order_id = od.order_id
JOIN Products p ON od.product_id = p.product_id
GROUP BY c.customer_id, p.product_id, p.product_name
ORDER BY c.customer_name, p.product_name;

-- 13. Total Revenue by Category
SELECT
    c.category_name,
    SUM(od.quantity * od.price) AS total_revenue
FROM Categories c
JOIN Products p ON c.category_id = p.category_id
JOIN OrderDetails od ON p.product_id = od.product_id
GROUP BY c.category_id, c.category_name
ORDER BY total_revenue DESC;

-- 14. Order Frequency by Month
SELECT
    strftime('%Y-%m', order_date) AS month,
    COUNT(order_id) AS order_count
FROM Orders
WHERE strftime('%Y', order_date) = '2023'
GROUP BY strftime('%Y-%m', order_date)
ORDER BY month;
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– 15. Products with Low Stock
SELECT
    p.product_name,
    SUM(od.quantity) AS total_quantity_ordered
FROM Products p
JOIN OrderDetails od ON p.product_id = od.product_id
GROUP BY p.product_id, p.product_name
HAVING SUM(od.quantity) < 10;
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